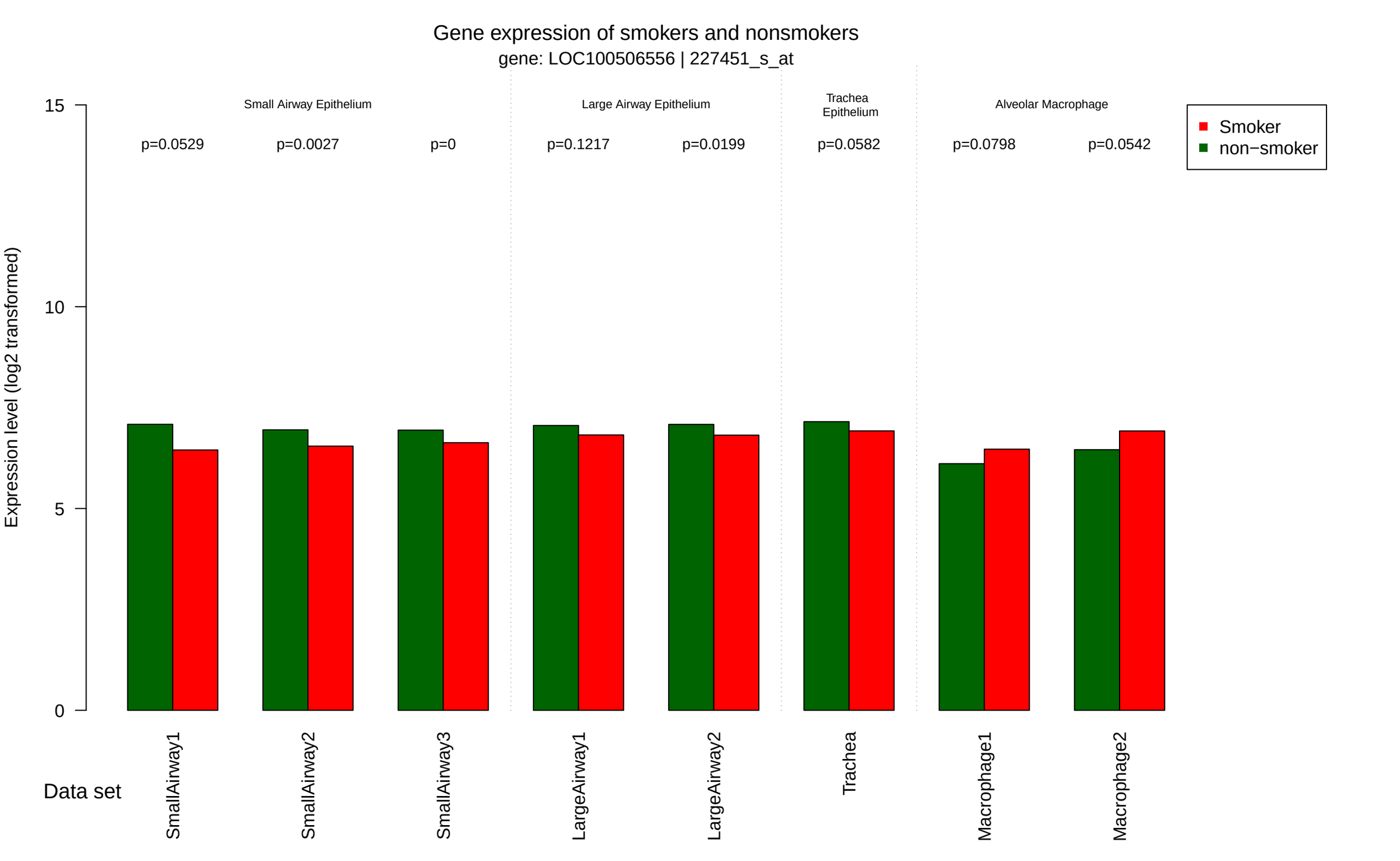
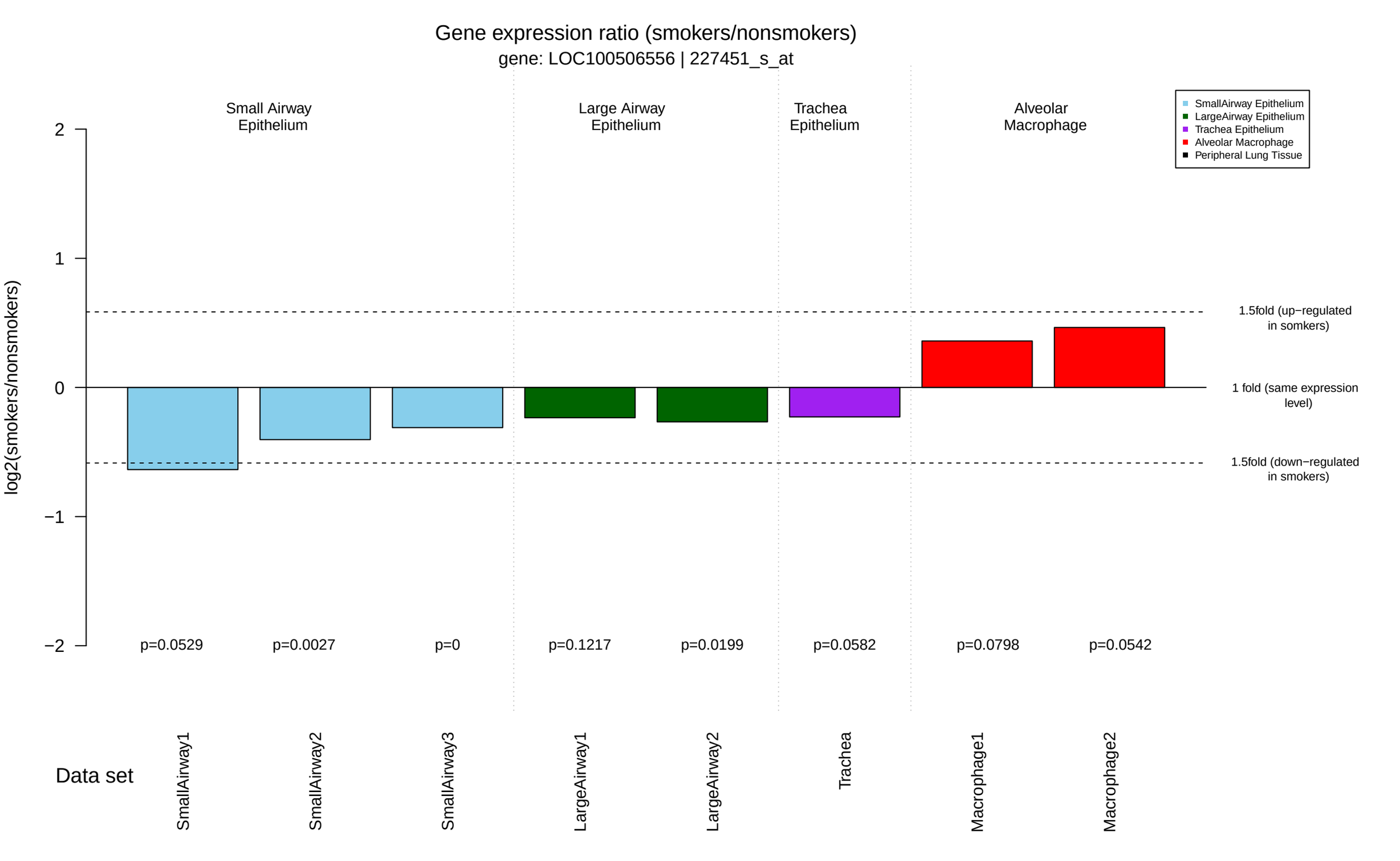
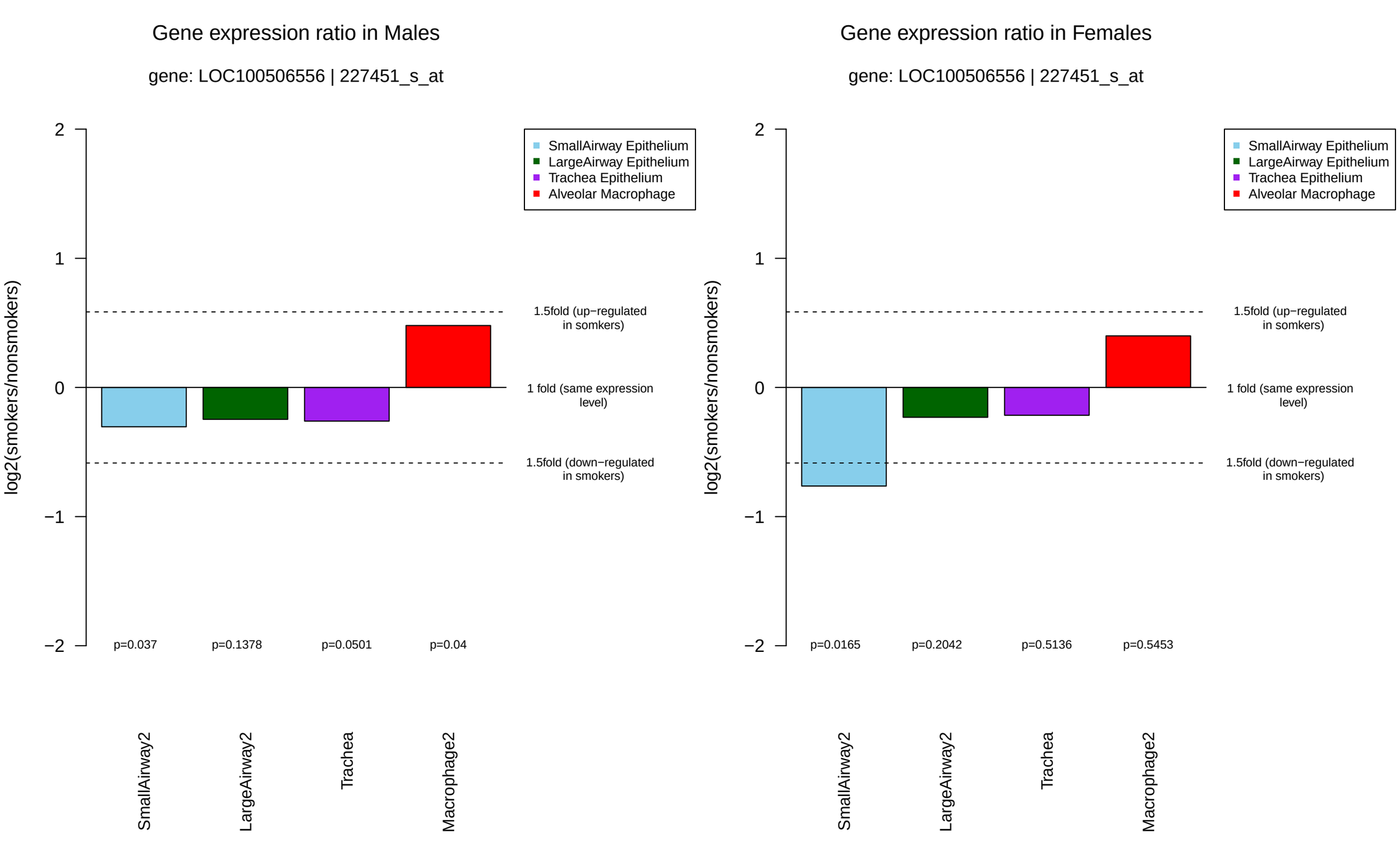
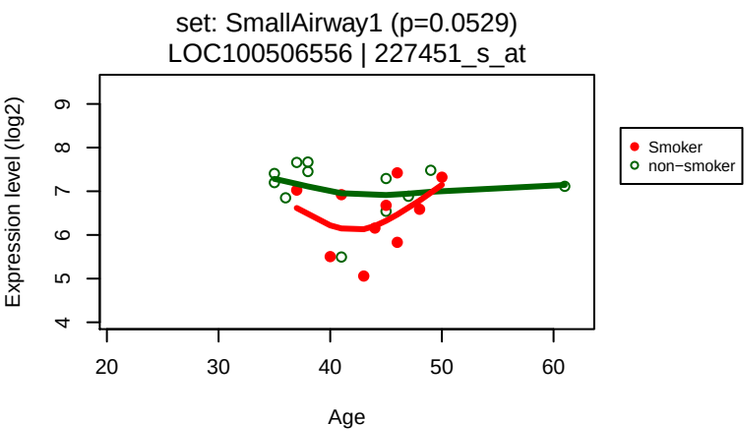
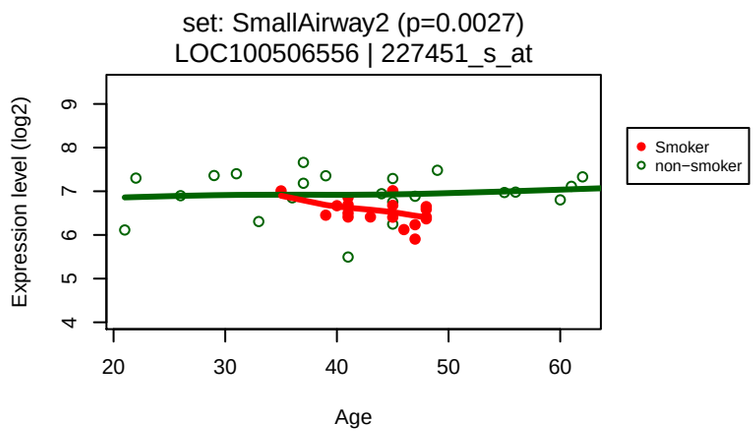
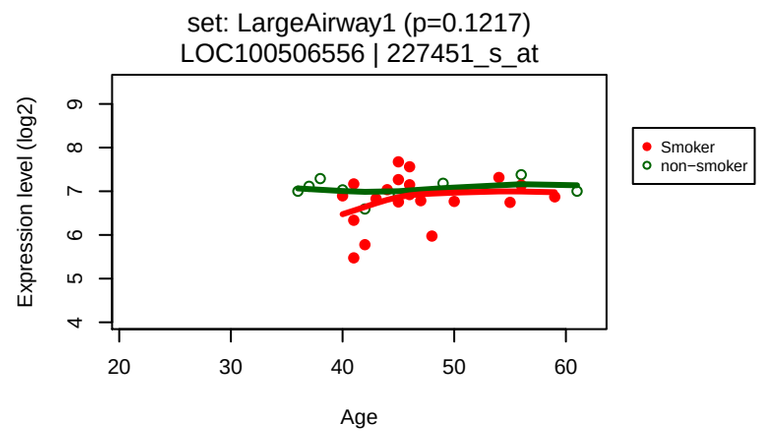

Gene expression of smokers and nonsmokers
gene: LOC100506556 | 227451_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0529
p=0.0027
p=0
p=0.1217
p=0.0199
p=0.0582
p=0.0798
p=0.0542
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

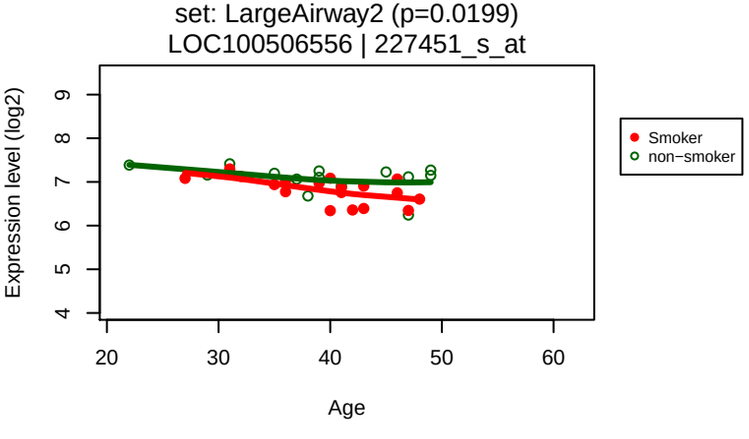

Gene expression ratio (smokers/nonsmokers)
gene: LOC100506556 | 227451_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0529
p=0.0027
p=0
p=0.1217
p=0.0199
p=0.0582
p=0.0798
p=0.0542
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

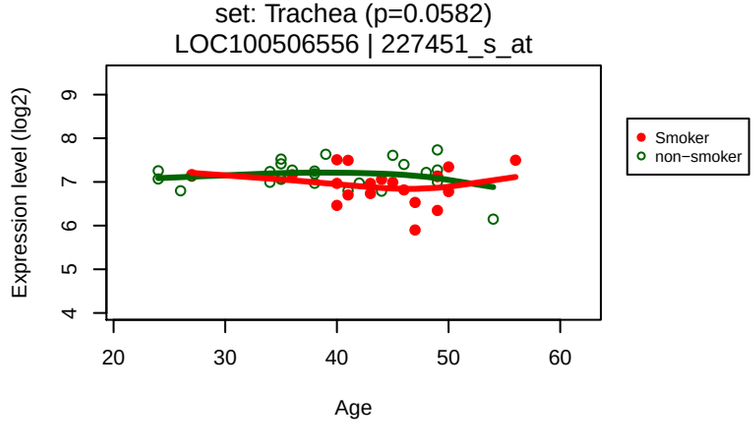

Gene expression ratio in Males
gene: LOC100506556 | 227451_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.037
p=0.1378
p=0.0501
p=0.04
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC100506556 | 227451_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0165
p=0.2042
p=0.5136
p=0.5453
SmallAirway2
LargeAirway2
Trachea
Macrophage2

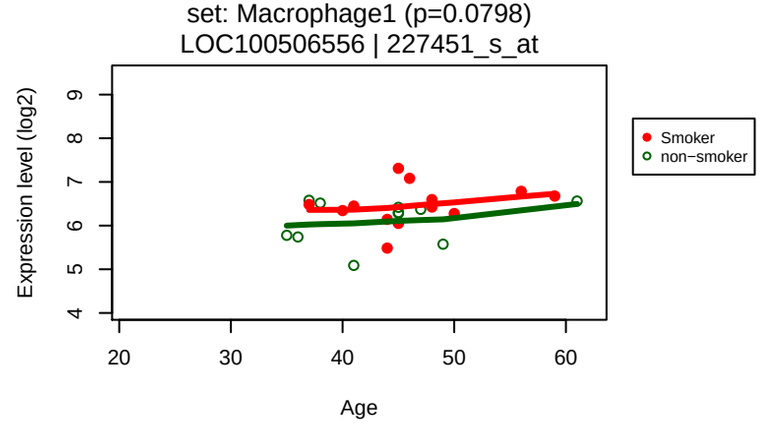

set: SmallAirway1 (p=0.0529)
LOC100506556 | 227451_s_at
Expression level (log2)
Age
Smoker
non−smoker

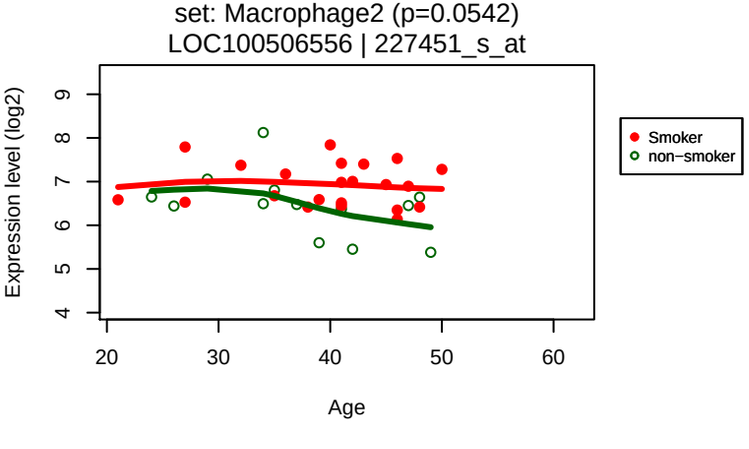

set: SmallAirway2 (p=0.0027)
LOC100506556 | 227451_s_at
Expression level (log2)
Age
Smoker
non−smoker

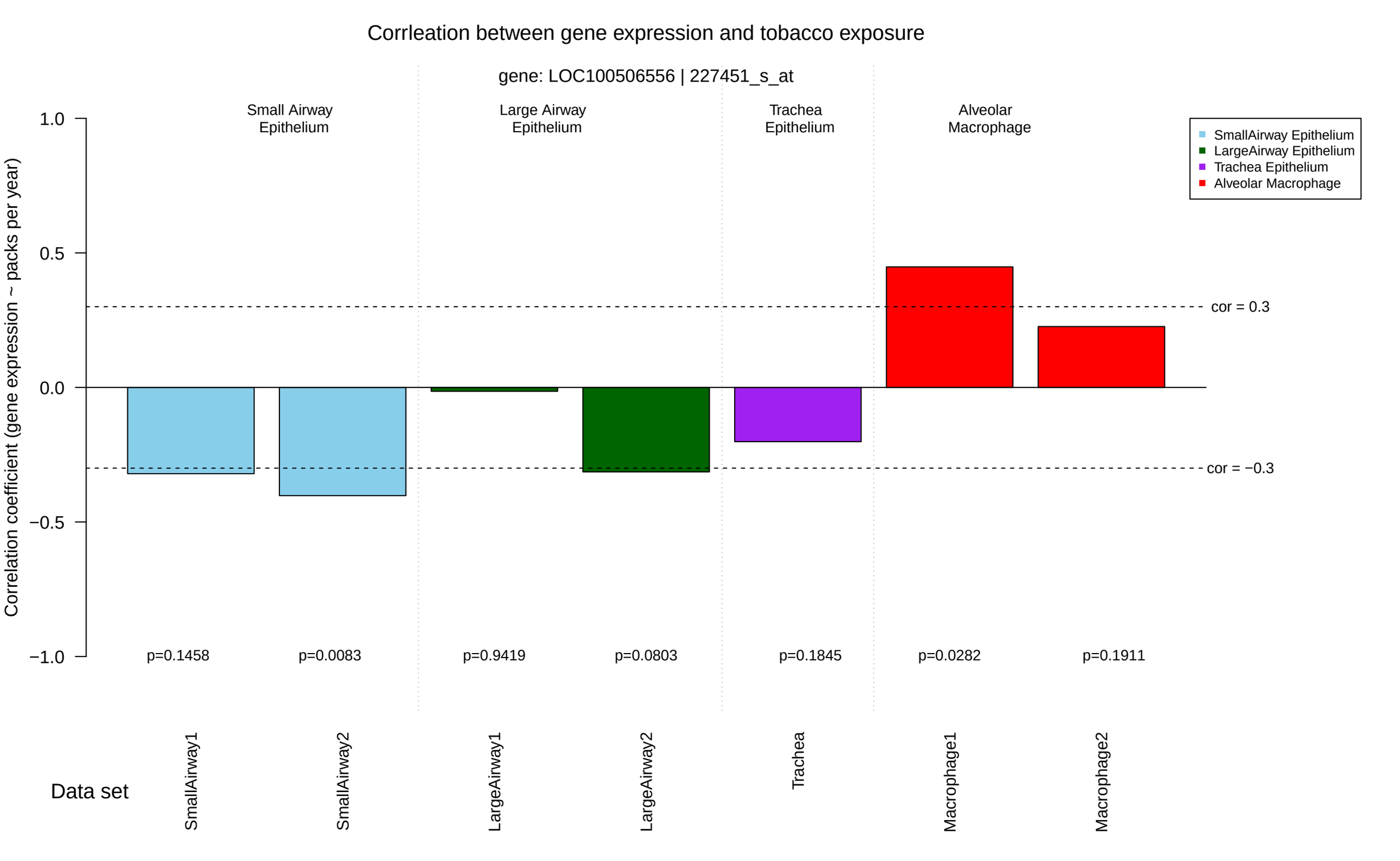

set: LargeAirway1 (p=0.1217)
LOC100506556 | 227451_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0199)
LOC100506556 | 227451_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0582)
LOC100506556 | 227451_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0798)
LOC100506556 | 227451_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0542)
LOC100506556 | 227451_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC100506556 | 227451_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.1458
p=0.0083
p=0.9419
p=0.0803
p=0.1845
p=0.0282
p=0.1911
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2